

DEEP LEARNING AND APPLICATIONS

MR20-1CS0158

UNIT I

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Machine Learning Basics: Learning Algorithms, Capacity, Over fitting, and Under fitting, Hyper parameters and Validation Sets, Estimators, Bias and Variance, Maximum Likelihood Estimation, Bayesian Statistics, Supervised and Unsupervised Learning algorithms, Stochastic Gradient Descent, Building a ML algorithm, Challenges and Motivation to Deep learning.

Text Book

1. Goodfellow, I., Bengio, Y., and Courville, A., Deep Learning, MIT Press, 2016

Machine Learning

- ❖ Machine learning is a branch of artificial intelligence that empowers computers to learn from data and improve their performance over time without explicit programming.
- ❖ It involves creating algorithms that can recognize patterns in data, make predictions, and solve complex tasks, leading to applications in areas such as image recognition, language processing, and autonomous systems.

Application:

- ❖ Develop a machine learning model that can analyze medical imaging data **to accurately detect and diagnose specific diseases, such as identifying early signs of diabetic retinopathy in retinal scans.**
- ❖ Construct a sentiment analysis model that can automatically determine the **sentiment (positive, negative, neutral) expressed in a given text, which could be used for analyzing social media posts or customer reviews.**
- ❖ Create a model that predicts stock market trends and prices by **analyzing historical market data and incorporating relevant economic indicators, assisting investors in making informed decisions.**

Deep Learning

- ❖ Deep learning is a subset of machine learning focused on neural networks with multiple layers, allowing automated extraction of complex data features.
- ❖ Deep learning is a method in artificial intelligence (AI) that teaches computers to process data in a way that is inspired by the human brain.
- ❖ **Deep learning models can recognize complex patterns in pictures, text, sounds, and other data to produce accurate insights and predictions.**
- ❖ It excels in tasks like image recognition, natural language processing, and autonomous systems, by learning hierarchical representations from data and eliminating the need for extensive manual feature engineering.

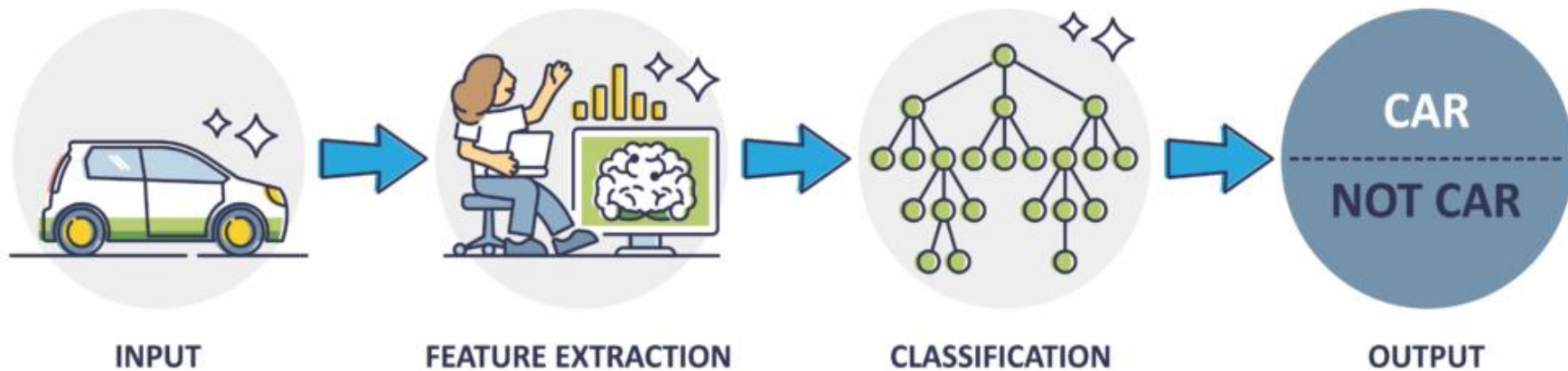
Application:

- ❖ Design a deep learning algorithm that enables a drone to navigate through complex environments and avoid obstacles, using onboard cameras for visual input.
- ❖ Create a deep learning model to segment and identify specific structures or anomalies in medical images, aiding in tasks like tumor detection in MRI scans.

Aspect	Machine Learning	Deep Learning
Definition	Subset of AI that learns patterns from data	Subset of ML using deep neural networks
Data Dependency	Works well with smaller, structured datasets	Requires large amounts of data, works with unstructured data
Feature Engineering	Requires manual feature extraction and selection	Automatically extracts features from raw data
Model Complexity	Simpler, more interpretable models	Complex models with multiple layers, harder to interpret
Computational Requirements	Less computational power needed	High computational power, often relies on GPUs
Applications	Recommendation systems, fraud detection, customer segmentation	Image/video analysis, speech recognition, natural language processing

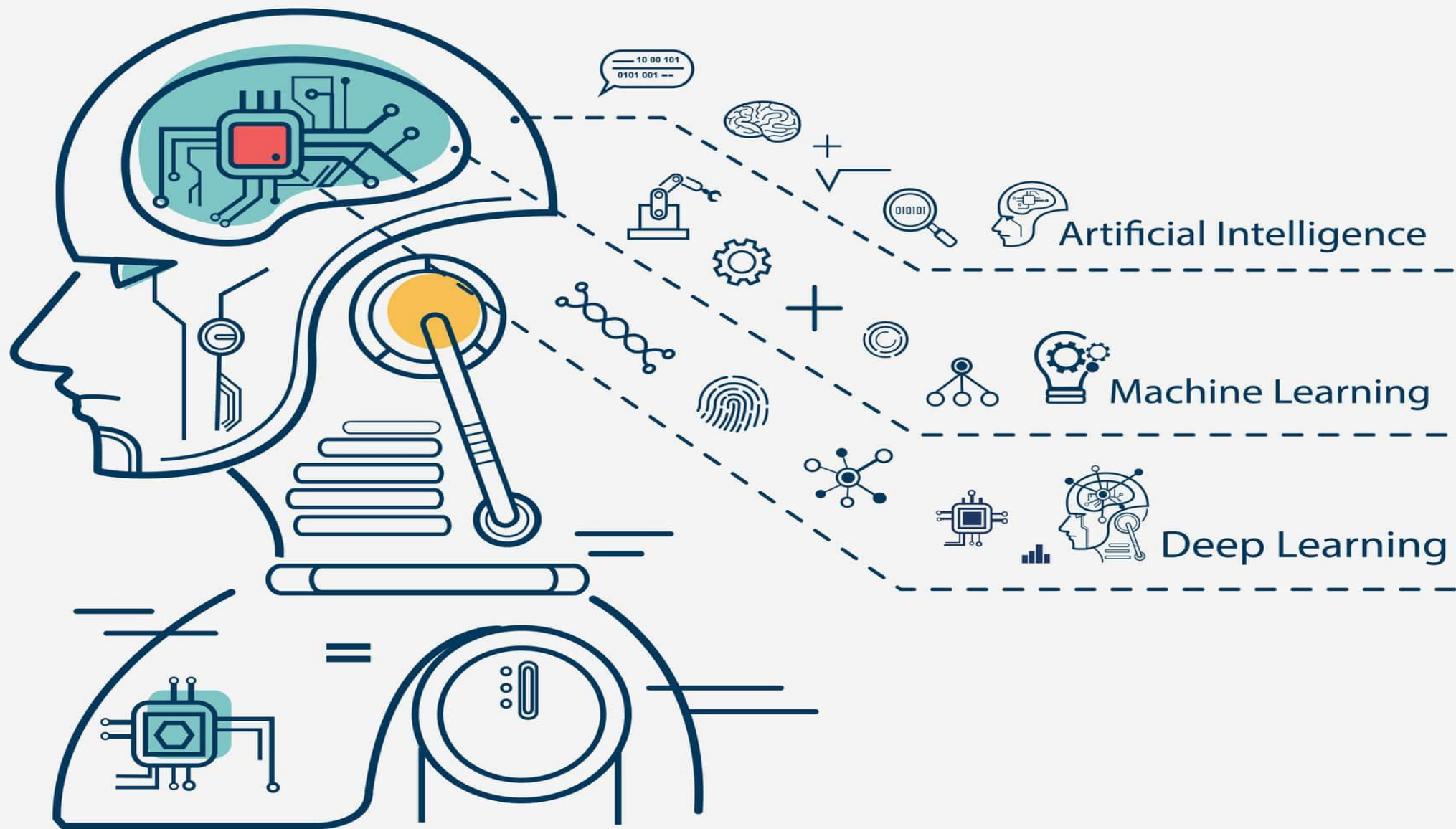
Aspect	Machine Learning	Deep Learning
Training Time	Shorter training times	Longer training times due to complexity and data volume
Example Algorithms	Linear regression, decision trees, SVM, k-NN	Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs)
Performance with Scale	May not improve significantly with more data	Performance improves with more data and deeper models
Interpretability	Generally more interpretable and explainable.	Often considered a "black box" with less interpretability.

MACHINE LEARNING



DEEP LEARNING





Learning Algorithms

- ❖ A machine learning algorithm is an algorithm that is able to learn from data.

What is the mean by learning?

- ❖ Mitchell (1997) provides the definition
- ❖ **“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .”**
- ❖ One can imagine a very wide variety of experiences E , tasks T , and performance measures P .
- ❖ Learning refers to the process by which a computer program (or system) becomes better at performing a certain task T as it gains more experience E .
- ❖ The improvement in performance is measured by a performance measure P . This process involves the program adjusting its internal parameters based on the data it encounters, allowing it to make better predictions or decisions over time.
- ❖ **The goal of learning is to enable the program to generalize from the provided data and perform well on new, unseen data.**

The Task, T

- ❖ Machine learning allows us to tackle tasks that are too difficult to solve with fixed programs written and designed by human beings.
- ❖ **From a scientific and philosophical point of view, machine learning is interesting because developing our understanding of the principles that underlie (behind) intelligence.**
- ❖ In this relatively formal definition of the word “task,”
- ❖ **“The process of learning itself is not the task. Learning is our means of attaining the ability to perform the task”.**
- ❖ For example, if we want a robot to be able to walk, then walking is the task.
- ❖ We could program the robot to learn to walk, or we could attempt to directly write a program that specifies how to walk manually.
- ❖ Machine learning tasks are usually described in terms of how the machine learning system should process.

- ❖ An example is a collection of features that have been quantitatively measured from some object or event that we want the machine learning system to process.
 - ❖ We typically represent an example as a vector $\mathbf{x} \in \mathbf{R}^n$ where each entry $\mathbf{x}_i$ of the vector is another feature.
 - ❖ For example, the features of an image are usually the values of the pixels in the image.
- Many kinds of tasks can be solved with machine learning. Some of the most common machine learning tasks include the following:**

1. Classification:

- ❖ Classification is a supervised machine learning method where the model tries to predict the correct label of a given input data.
- ❖ In classification, the model is fully trained using the training data, and then it is evaluated on test data before being used to perform prediction on new unseen data.

- ❖ In this type of task, the computer program is asked to specify which of k categories some input belongs to.
- ❖ To solve this task, the learning algorithm is usually asked to produce a function
$$\mathbf{f} : \mathbf{R}^n \rightarrow \{1, \dots, k\}.$$
- ❖ When $\mathbf{y} = \mathbf{f}(\mathbf{x})$, the model assigns an input described by vector $\mathbf{x}$ to a category identified by numeric code y .
- ❖ There are other variants of the classification task, for example, where f outputs a probability distribution over classes.
- ❖ An example of a classification task is object recognition, where the input is an image (usually described as a set of pixel brightness values), and the output is a numeric code identifying the object in the image.

2. Classification with missing inputs:

- ❖ Classification becomes more challenging if the computer program is not guaranteed that every measurement in its input vector will always be provided.
- ❖ In order to solve the classification task, the learning algorithm only has to define a single function mapping from a vector input to a categorical output.
- ❖ When some of the inputs may be missing, rather than providing a single classification function, the learning algorithm must learn a set of functions. Each function corresponds to classifying x with a different subset of its inputs missing.
- ❖ This kind of situation arises frequently in medical diagnosis, because many kinds of medical tests are expensive or invasive.
- ❖ One way to efficiently define such a large set of functions is to learn a probability distribution over all of the relevant variables, then solve the classification task by marginalizing out the missing variables.

3. Regression:

- ❖ **Regression is a supervised machine learning technique which is used to predict continuous values. The ultimate goal of the regression algorithm is to plot a best-fit line or a curve between the data. The three main metrics that are used for evaluating the trained regression model are variance, bias and error.**
- ❖ In this type of task, the computer program is asked to predict a numerical value given some input.
- ❖ To solve this task, the learning algorithm is asked to output a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$.
- ❖ This type of task is similar to classification, except that the format of output is different.
- ❖ An example of a regression task is the prediction of the expected claim amount that an insured person will make (used to set insurance premiums), or the prediction of future prices of securities.
- ❖ These kinds of predictions are also used for algorithmic trading.

4. Transcription:

- ❖ In this type of task, the machine learning system is asked to observe a relatively unstructured representation of some kind of data and transcribe it into discrete, textual form.
- ❖ For example, in optical character recognition, the computer program is shown a photograph containing an image of text and is asked to return this text in the form of a sequence of characters (e.g., in ASCII or Unicode format).

5. Machine translation:

- ❖ In a machine translation task, the input already consists of a sequence of symbols in some language, and the computer program must convert this into a sequence of symbols in another language.
- ❖ This is commonly applied to natural languages, such as to translate from English to French.
- ❖ Deep learning has recently begun to have an important impact on this kind of task (Sutskever et al., 2014; Bahdanau et al., 2015).

6. Structured output:

- ❖ Structured output tasks involve any task where the output is a vector (or other data structure containing multiple values) with important relationships between the different elements.
- ❖ This is a broad category, and incorporates the transcription and translation tasks described above, but also many other tasks.
- ❖ One example is parsing—mapping a natural language sentence into a tree that describes its grammatical structure and tagging nodes of the trees as being verbs, nouns, or adverbs, and so on.

7. Anomaly detection:

- ❖ In this type of task, the computer program sifts through a set of events or objects, and flags some of them as being unusual or uncharacteristic.
- ❖ An example of an anomaly detection task is credit card fraud detection.

- ❖ By modeling your purchasing habits, a credit card company can detect misuse of your cards.
- ❖ If a thief steals your credit card or credit card information, the thief's purchases will often come from a different probability distribution over purchase types than your own.
- ❖ The credit card company can prevent fraud by placing a hold on an account as soon as that card has been used for an uncharacteristic purchase.

8. Synthesis and sampling:

- ❖ In this type of task, the machine learning algorithm is asked to generate new examples that are similar to those in the training data.
- ❖ Synthesis and sampling via machine learning can be useful for media applications where it can be expensive or boring for an artist to generate large volumes of content by hand.
- ❖ For example, video games can automatically generate textures for large objects or landscapes, rather than requiring an artist to manually label each pixel

9. Imputation of missing values:

- ❖ In this type of task, the machine learning algorithm is given a new example $x \in \mathbb{R}^n$, but with some entries x_i of x missing.
- ❖ The algorithm must provide a prediction of the values of the missing entries.

10. Denoising:

- ❖ In this type of task, the machine learning algorithm is given in input a corrupted example $\tilde{x} \in \mathbb{R}^n$ obtained by an unknown corruption process from a clean example $x \in \mathbb{R}^n$.
- ❖ The learner must predict the clean example x from its corrupted version $\tilde{x}$, or more generally predict the conditional probability distribution $p(x | \tilde{x})$.

11. Density estimation or probability mass function estimation:

- ❖ In the density estimation problem, the machine learning algorithm is asked to learn a function $P_{\text{model}} : \mathbb{R}^n \rightarrow \mathbb{R}$, where $P_{\text{model}}(x)$ can be interpreted as a probability density function (if x is continuous) or a probability mass function (if x is discrete) on the space that the examples were drawn from.
- ❖ To do such a task well (*we will specify exactly what that means when we discuss performance measures P*), the algorithm needs to learn the structure of the data it has seen. It must know where examples cluster tightly and where they are unlikely to occur.
- ❖ Most of the tasks described above require that the learning algorithm has at least implicitly captured the structure of the probability distribution.
- ❖ Of course, many other tasks and types of tasks are possible. The types of tasks we list here are intended only to provide examples of what machine learning can do, not to define a rigid taxonomy (classification) of tasks.

❖ The Performance Measure, P

- ❖ In order to evaluate the abilities of a machine learning algorithm, we must design a quantitative measure of its performance.
- ❖ Usually this performance measure P is specific to the task T being carried out by the system.
- ❖ **For tasks such as classification, classification with missing inputs, and transcription, we often measure the accuracy of the model.**
- ❖ Accuracy is just the proportion of **examples for which the model produces the correct output**. We can also obtain equivalent information by measuring the error rate, the proportion of examples for which **the model produces an incorrect output**.
- ❖ We often refer to the error rate as the expected 0-1 loss. The 0-1 loss on a particular example is 0 if it is correctly classified and 1 if it is not.
- ❖ For tasks such as density estimation, it does not make sense to measure accuracy, error rate, or any other kind of 0-1 loss.

- ❖ Instead, we must use a different performance metric that gives the model a continuous-valued score for each example. The most common approach is to report the **average log-probability the model assigns to some examples**
- ❖ The choice of performance measure may seem straightforward and objective, but it is often difficult to choose a performance measure that corresponds well to the desired behavior of the system.
- ❖ In some cases, this is because it is difficult to decide what should be measured.
- ❖ For example, when performing a transcription task, should we measure the accuracy of the system at transcribing entire sequences, or should we use a more fine-grained performance measure that gives partial credit for getting some elements of the sequence correct?
- ❖ When performing a regression task, should we correct the system more if it frequently makes medium-sized mistakes or if it rarely makes very large mistakes?
- ❖ **These kinds of design choices depend on the application.**

The Experience, E

- ❖ Machine learning algorithms can be broadly categorized as unsupervised or supervised by what kind of experience they are allowed to have during the learning process.
- ❖ Most of the learning algorithms can be understood as being allowed to experience an entire dataset.
- ❖ One of the oldest datasets studied by statisticians and machine learning researchers is the Iris dataset (Fisher, 1936).
- ❖ It is a collection of measurements of different parts of 150 iris plants. Each individual plant corresponds to one example.
- ❖ The features within each example are the measurements of each of the parts of the plant: the **sepal length, sepal width, petal length and petal width**.
- ❖ The dataset also records which species each plant belonged to. Three different species are represented in the dataset.

- ❖ Unsupervised learning algorithms experience a dataset containing many features, then learn useful properties of the structure of this dataset.
- ❖ In the context of deep learning, we usually want to learn the entire probability distribution that generated a dataset, whether explicitly as in density estimation or implicitly for tasks like synthesis or denoising.
- ❖ **Some other unsupervised learning algorithms perform other roles, like clustering, which consists of dividing the dataset into clusters of similar examples.**
- ❖ **Supervised learning algorithms experience a dataset containing features, but each example is also associated with a label or target.**
- ❖ For example, the Iris dataset is annotated with the species of each iris plant.
- ❖ A supervised learning algorithm can study the Iris dataset and learn to classify iris plants into three different species based on their measurements.

- ❖ **Some machine learning algorithms do not just experience a fixed dataset. For example, reinforcement learning algorithms interact with an environment, so there is a feedback loop between the learning system and its experiences.**
- ❖ **Most machine learning algorithms simply experience a dataset. A dataset can be described in many ways. In all cases, a dataset is a collection of examples, which are in turn collections of features.**
- ❖ **There is no formal definition of supervised and unsupervised learning, there is no rigid taxonomy of datasets or experiences. The structures described here cover most cases, but it is always possible to design new ones for new applications.**

Capacity, Overfitting and Underfitting

- ❖ The central challenge in machine learning is that we must perform well on new, previously unseen inputs—not just those on which our model was trained.
- ❖ **The ability to perform well on previously unobserved inputs is called generalization.**
- ❖ Typically, when training a machine learning model, we have access to a training set, we can compute some error measure on the training set called the **training error**, and we reduce this training error.
- ❖ So far, what we have described is simply an optimization problem.
- ❖ What separates machine learning from optimization is that we want the generalization error, also called the test error, to be low as well.

- ❖ **The generalization error is defined as the expected value of the error on a new input.**
- ❖ Here the expectation is taken across different possible inputs, drawn from the distribution of inputs we expect the system to encounter in practice.
- ❖ We typically estimate the generalization error of a machine learning model by measuring its performance on a test set of examples that were collected separately from the training set.

In our linear regression example, we trained the model by minimizing the training error,

$$\frac{1}{m^{(\text{train})}} \| \mathbf{X}^{(\text{train})} \mathbf{w} - \mathbf{y}^{(\text{train})} \|_2^2, \quad (5.14)$$

but we actually care about the test error, $\frac{1}{m^{(\text{test})}} \| \mathbf{X}^{(\text{test})} \mathbf{w} - \mathbf{y}^{(\text{test})} \|_2^2$.

- ❖ If the training and the test set are collected randomly.
- ❖ If we are allowed to make some assumptions about how the training and test set are collected, then we can make some progress.
- ❖ **The train and test data are generated by a probability distribution over datasets called the data generating process.**
- ❖ We typically make a set of assumptions known collectively as the these assumptions **i.i.d. assumptions (Independent and identically distributed random variables)** are that the examples in each dataset are independent from each other, and that the train set and test set are identically distributed, drawn from the same probability distribution as each other.
- ❖ This assumption allows us to describe the data generating process with a probability distribution over a single example.
- ❖ The same distribution is then used to generate every train example and every test example.
- ❖ **We call that shared underlying distribution the data generating distribution, denoted p_{data} .**

- ❖ This **probabilistic framework and the i.i.d. assumptions** (Independent and identically distributed random variables) allow us to mathematically study the relationship between training error and test error.
- ❖ **One immediate connection we can observe between the training and test error is that the expected training error of a randomly selected model is equal to the expected test error of that model.**
- ❖ Suppose we have a probability distribution $p(x, y)$ and we sample from it repeatedly to generate the train set and the test set.
- ❖ For some fixed value w , the expected training set error is exactly the same as the expected test set error, because both expectations are formed using the same dataset sampling process.
- ❖ The only difference between the two conditions is the name we assign to the dataset we sample.

- ❖ We sample the training set, then use it to choose the parameters to reduce training set error, then sample the test set.
- ❖ Under this process, the expected test error is greater than or equal to the expected value of training error.
- ❖ The factors determining how well a machine learning algorithm will perform are its ability to:
 1. **Make the training error small.**
 2. **Make the gap between training and test error small.**
- ❖ These two factors correspond to the two central challenges in machine learning: underfitting and overfitting.
- ❖ Underfitting occurs when the model is not able to obtain a sufficiently low error value on the training set.
- ❖ Overfitting occurs when the gap between the training error and test error is too large.
- ❖ We can control whether a model is more likely to overfit or underfit by altering its capacity. Informally, a **model's capacity is its ability to fit a wide variety of functions.**

- ❖ **Models with low capacity may struggle to fit the training set.**
- ❖ Models with high capacity can overfit by memorizing properties of the training set that do not serve them well on the test set.
- ❖ One way to control the capacity of a learning algorithm is by choosing its hypothesis space, the set of functions that the learning algorithm is allowed to select as being the solution.
- ❖ For example, the linear regression algorithm has the set of all linear functions of its input as its hypothesis space.
- ❖ We can generalize linear regression to include polynomials, rather than just linear functions, in its hypothesis space.
- ❖ Doing so increases the model's capacity.
- ❖ A polynomial of degree one gives us the linear regression model with which we are already familiar, with prediction
$$\hat{y} = b + wx.$$

- ❖ By introducing x_2 as another feature provided to the linear regression model, we can learn a model that is quadratic as a function of x :

$$\hat{y} = b + w_1x_1 + w_2x_2$$

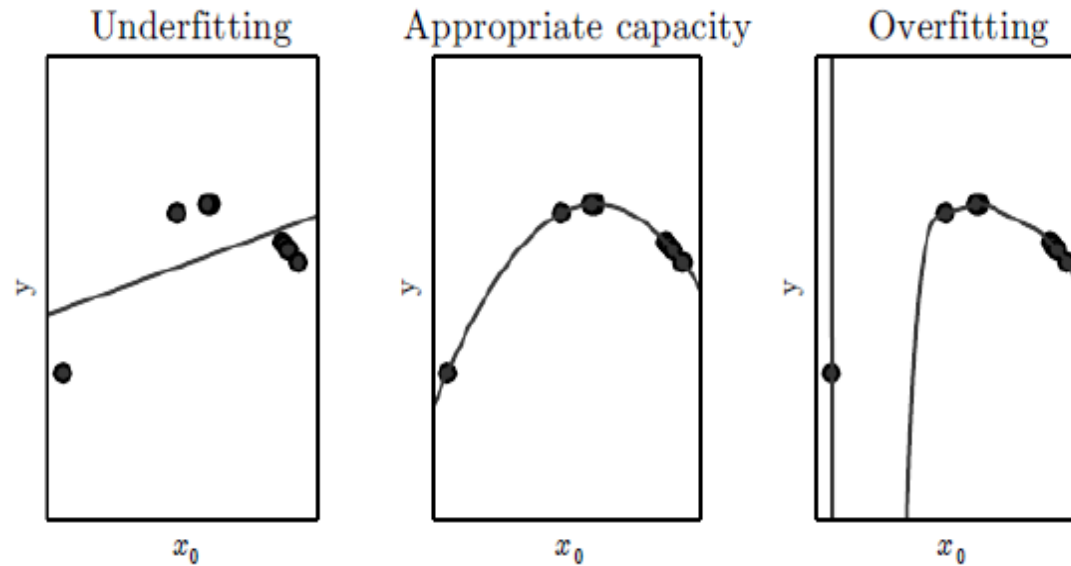
- ❖ Though this model implements a quadratic function of its input, the output is still a linear function of the parameters, so we can still use the normal equations to train the model in closed form. We can continue to add more powers of x as additional features, for example to obtain a polynomial of degree 9:

$$\hat{y} = b + \sum_{i=1}^9 w_i x^i.$$

- ❖ **Machine learning algorithms will generally perform best when their capacity is appropriate in regard to the true complexity of the task they need to perform and the amount of training data they are provided with.**
- ❖ Models with insufficient capacity are unable to solve complex tasks. Models with high capacity can solve complex tasks, but when their capacity is higher than needed to solve the present task they may overfit.

- ❖ **Fig. 5.2** shows this principle in action. We compare a linear, quadratic and degree-9 predictor attempting to fit a problem where the true underlying function is quadratic.
- ❖ **The linear function is unable to capture the curvature in the true underlying problem, so it underfits.**
- ❖ The degree-9 predictor is capable of representing the correct function, but it is also capable of representing infinitely many other functions that pass exactly through the training points, because we have more parameters than training examples.
- ❖ We have little chance of choosing a solution that generalizes well when so many wildly different solutions exist.
- ❖ In this example, the quadratic model is perfectly matched to the true structure of the task so it generalizes well to new data.
- ❖ **Figure 5.2:** We fit three models to this example training set. The training data was generated synthetically, by randomly sampling x values and choosing y deterministically by evaluating a quadratic function.

Figure 5.2:



- ❖ (Left) A linear function fit to the data suffers from underfitting—it cannot capture the curvature that is present in the data.
- ❖ (Center) A quadratic function fit to the data generalizes well to unseen points. It does not suffer from a significant amount of overfitting or underfitting.
- ❖ (Right) A polynomial of degree 9 fit to the data suffers from overfitting. Here we used the Moore-Penrose pseudoinverse to solve the underdetermined normal equations.

- ❖ The solution passes through all of the training points exactly, but we have not been lucky enough for it to extract the correct structure.
- ❖ It now has a deep valley in between two training points that does not appear in the true underlying function.
- ❖ It also increases sharply on the left side of the data, while the true function decreases in this area.
- ❖ So far we have only described changing a model's capacity by changing the number of input features it has (and simultaneously adding new parameters associated with those features).
- ❖ There are in fact many ways of changing a model's capacity. Capacity is not determined only by the choice of model.

- ❖ The model specifies which family of functions the learning algorithm can choose from when varying the parameters in order to reduce a training objective.
- ❖ This is called the representational capacity of the model. In many cases, finding the best function within this family is a very difficult optimization problem.
- ❖ In practice, the learning algorithm does not actually find the best function, but merely one that significantly reduces the training error.
- ❖ These additional limitations, such as the imperfection of the optimization algorithm, mean that the learning algorithm's effective capacity may be less than the representational capacity of the model family

- ❖ Figure 5.3: Typical relationship between capacity and error. Training and test error behave differently. **At the left end of the graph, training error and generalization error are both high.** This is the underfitting rule.
- ❖ As we **increase capacity, training error decreases, but the gap between training and generalization error increases.** Eventually, the size of this gap outweighs the decrease in training error, and we enter the overfitting rule, where capacity is too large, above the optimal capacity.

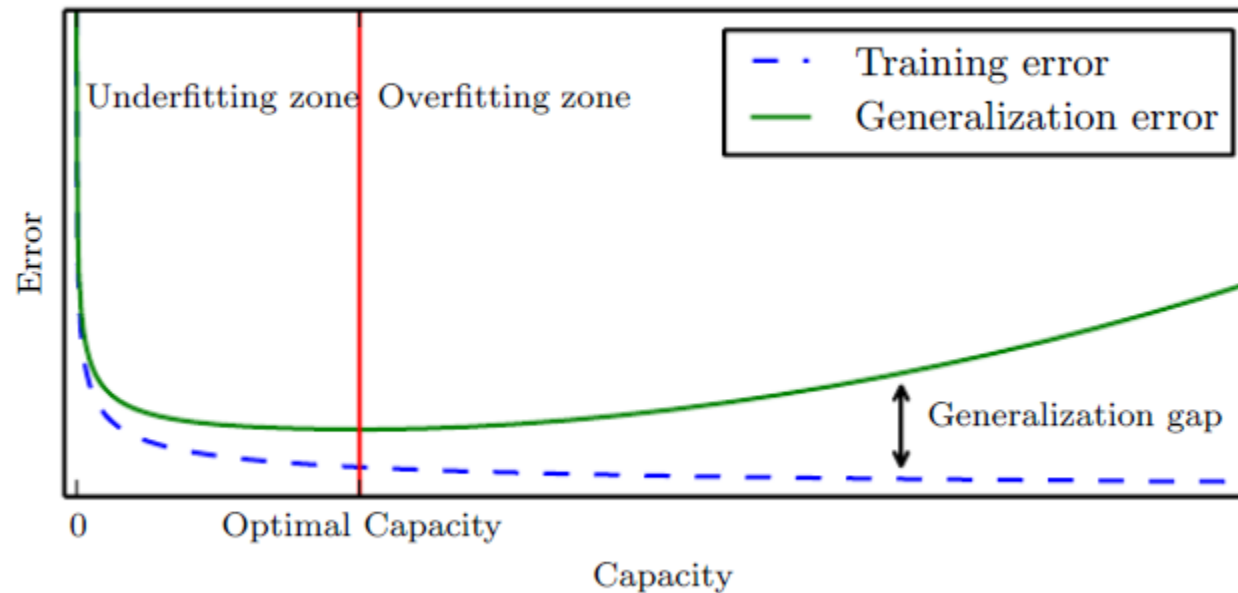


Figure 5.3

❖ Regularization

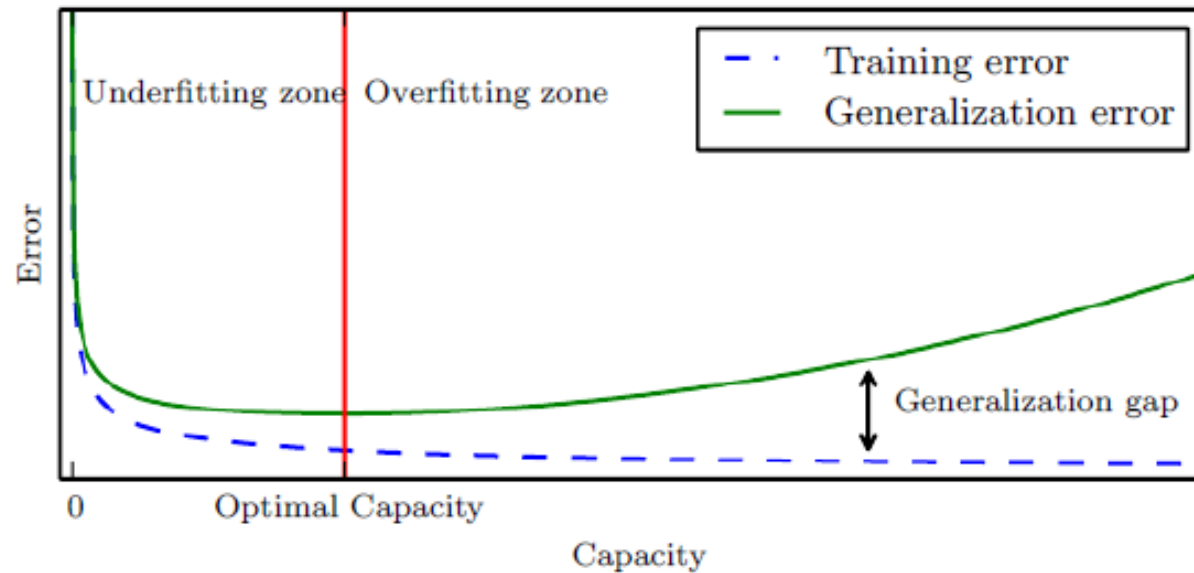
- ❖ The **no free lunch theorem** implies that we must design our machine learning algorithms to perform well on a specific task. We do so by building a set of preferences into the learning algorithm. When these preferences are aligned with the learning problems we ask the algorithm to solve, it performs better.
- ❖ Regularization is a technique used in machine learning and deep learning to reduce overfitting by discouraging the model from becoming too complex.
- ❖ It does this by adding a penalty to the loss function, which limits how much the model can rely on large weights or complicated structures.
- ❖ In simple terms, regularization helps the model focus on the most important patterns in the training data and ignore noise or random fluctuations, leading to better performance on new, unseen data.
- ❖ There are different types of regularization:
 - ❖ L1 regularization (Lasso): Makes some weights exactly zero, encouraging sparsity.
 - ❖ L2 regularization (Ridge or weight decay): Keeps weights small, encouraging smoother models.

- ❖ The "No Free Lunch" (NFL) theorem in optimization and machine learning states that no single optimization algorithm or machine learning model can universally outperform all others across all possible problems. Essentially, if an algorithm performs well on a specific set of problems, it must perform correspondingly worse on others when averaged over all possible problems. This implies that there is no "magic bullet" algorithm that works best in all situations.
- ❖ Regularization is any modification we make to a learning algorithm that is intended to reduce its generalization error but not its training error.
- ❖ Regularization is one of the central concerns of the field of machine learning, rivaled in its importance only by optimization.
- ❖ The no free lunch theorem has made it clear that there is no best machine learning algorithm, and, in particular, no best form of regularization. Instead we must choose a form of regularization that is well-suited to the particular task we want to solve.

Hyperparameters and Validation Sets

- ❖ **Most machine learning algorithms have several settings that we can use to control the behavior of the learning algorithm. These settings are called hyperparameters.**
- ❖ **The values of hyperparameters are not adapted by the learning algorithm itself (though we can design a nested learning procedure where one learning algorithm learns the best hyperparameters for another learning algorithm).**
- ❖ **Hyperparameters are parameters whose values control the learning process and determine the values of model parameters that a learning algorithm ends up learning. The prefix 'hyper_' suggests that they are 'top-level' parameters that control the learning process and the model parameters that result from it.**
- ❖ **What is the difference between parameters and hyperparameters?**
- ❖ **Parameters are the internal values (like weights and biases in a neural network) that are automatically learned from the training data and directly determine the model's output.**
- ❖ **Hyperparameters are the external settings (like learning rate, batch size, number of layers) manually defined before training that guide the learning process but are not updated during it.**

- ❖ Sometimes a setting is chosen to be a hyperparameter that the learning algorithm does not learn because it is difficult to optimize
- ❖ More frequently, we do not learn the hyperparameter because it is not appropriate to learn that hyperparameter on the training set.
- ❖ This applies to **all hyperparameters that control model capacity**. If learned on the training set, such hyperparameters would always choose the maximum possible model capacity, resulting in overfitting (refer to Fig.).



- ❖ **For example**, we can always fit the training set better with a higher degree polynomial and a weight decay setting of $\lambda = 0$ than we could with a lower degree polynomial and a positive weight decay setting.
- ❖ To solve this problem, **we need a validation set of examples that the training algorithm does not observe.**
- ❖ Earlier we discussed how a held-out test set, composed of examples coming from the same distribution as the training set, can be used to estimate the generalization error of a learner, after the learning process has completed.
- ❖ It is important that the test examples are not used in any way to make choices about the model, including its hyperparameters.
- ❖ For this reason, no example from the test set can be used in the validation set. Therefore, we always construct the validation set from the training data.
- ❖ Specifically, we split the training data into two disjoint subsets.

- ❖ One of these subsets is used to learn the parameters. The other subset is our validation set, used to estimate the generalization error during or after training, allowing for the hyperparameters to be updated accordingly.
- ❖ The subset of data used to learn the parameters is still typically called the training set, even though this may be confused with the larger pool of data used for the entire training process.
- ❖ **The subset of data used to guide the selection of hyperparameters is called the validation set.**
- ❖ Typically, one uses about 80% of the training data for training and 20% for validation.
- ❖ Since the validation set is used to “train” the hyperparameters, the validation set error will underestimate the generalization error, though typically by a smaller amount than the training error.
- ❖ After all hyperparameter optimization is complete, the generalization error may be estimated using the test set.

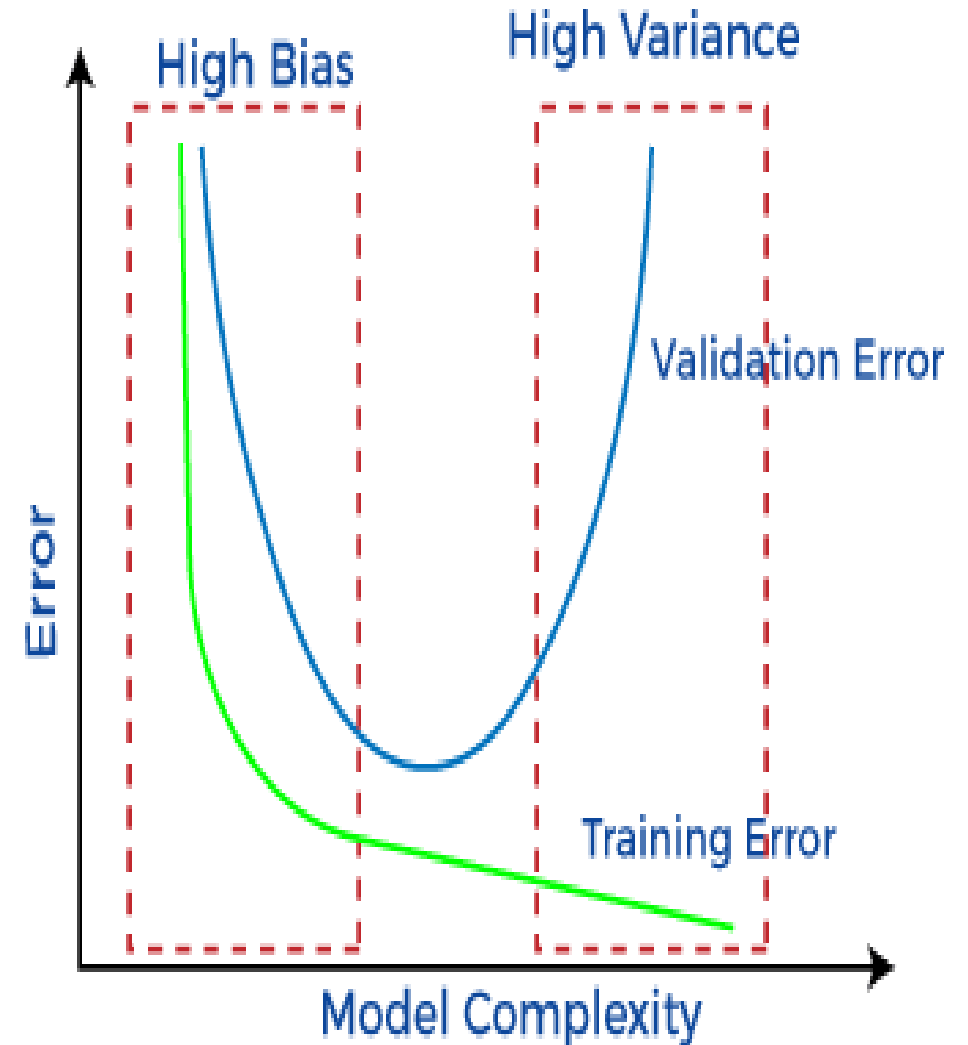
Cross-Validation in ML

- ❖ **Cross validation is a technique used in machine learning to evaluate the performance of a model on unseen data.**
- ❖ **It involves dividing the available data into multiple folds or subsets, using one of these folds as a validation set, and training the model on the remaining folds.**
- ❖ **This process is repeated multiple times, each time using a different fold as the validation set.**
- ❖ **Finally, the results from each validation step are averaged to produce a more robust estimate of the model's performance.**
- ❖ **Cross validation is an important step in the machine learning process and helps to ensure that the model selected for deployment is robust and generalizes well to new data.**

- ❖ **Cross-validation is a technique for evaluating ML models by training several ML models on subsets of the available input data and evaluating them on the complementary subset of the data. Use cross-validation to detect overfitting, ie, failing to generalize a pattern.**
- ❖ **Why do we need cross-validation?**
- ❖ Cross Validation is commonly used in Machine Learning to compare different models and select the most appropriate one for a specific problem. It is both easy to understand, easy to implement, and less biased than other methods.
- ❖ **Purpose of cross validation in machine learning**
- ❖ **The purpose of cross-validation is to test the ability of a machine learning model to predict new data.**
- ❖ **It is also used to flag problems like overfitting or selection bias and gives insights on how the model will generalize to an independent dataset.**

❖ Bias and Variance in Machine Learning

- ❖ Machine learning is a branch of Artificial Intelligence, which allows machines to perform data analysis and make predictions.
- ❖ However, if the machine learning model is not accurate, it can make predictions errors, and these prediction errors are usually known as **Bias and Variance**.
- ❖ In machine learning, these errors will always be present as there is always a slight difference between the model predictions and actual predictions.
- ❖ The main aim of ML/data science analysts is to reduce these errors in order to get more accurate results.

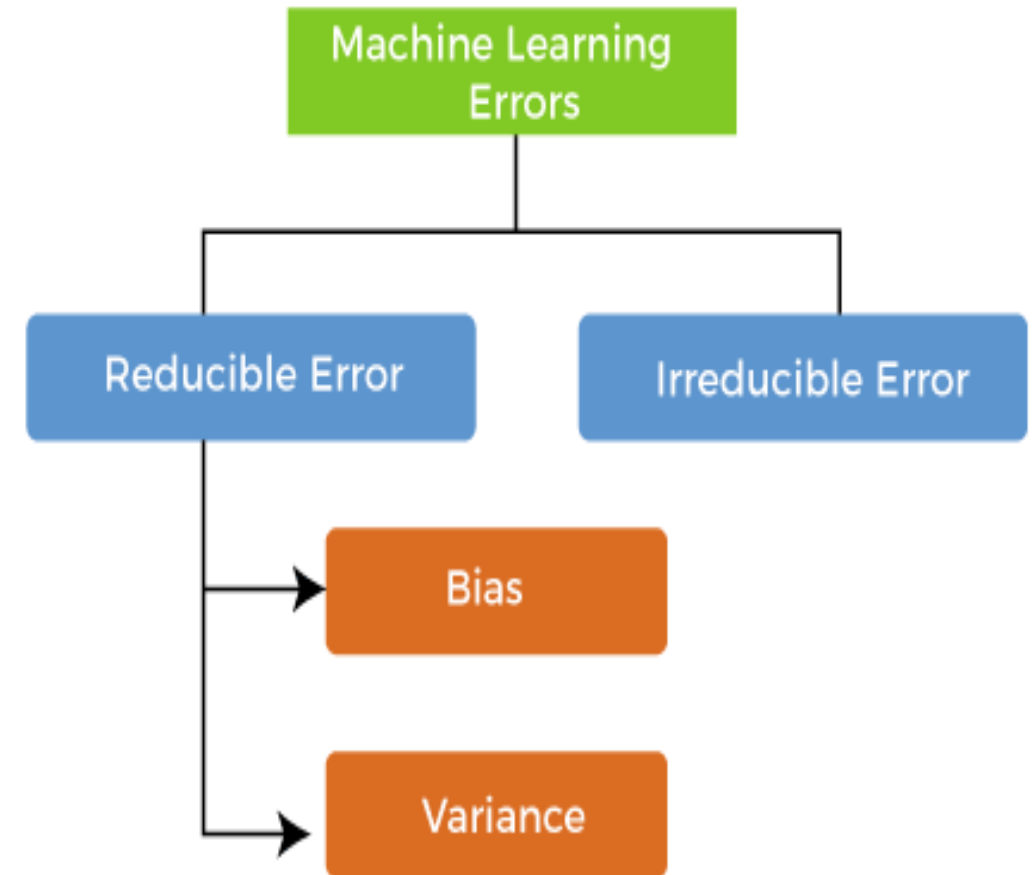


❖ Errors in Machine Learning?

❖ In machine learning, an error is a measure of how accurately an algorithm can make predictions for the previously unknown dataset. On the basis of these errors, the machine learning model is selected that can perform best on the particular dataset. There are mainly two types of errors in machine learning, which are:

❖ **Reducible errors:** These errors can be reduced to improve the model accuracy. Such errors can further be classified into bias and Variance.

❖ **Irreducible errors:** These errors will always be present in the model regardless of which algorithm has been used. The cause of these errors is unknown variables whose value can't be reduced.



❖ What is Bias?

- ❖ In general, a machine learning model analyses the data, find patterns in it and make predictions.
- ❖ While training, the model learns these patterns in the dataset and applies them to test data for prediction.
- ❖ **While making predictions, a difference occurs between prediction values made by the model and actual values and expected values, and this difference is known as bias errors or Errors due to bias.**
- ❖ It can be defined as an inability of machine learning algorithms such as Linear Regression to capture the true relationship between the data points.
- ❖ Each algorithm begins with some amount of bias because bias occurs from assumptions in the model, which makes the target function simple to learn.

- ❖ A model has either:
- ❖ **Low Bias:** A low bias model will make fewer assumptions about the form of the target function.
- ❖ **High Bias:** A model with a high bias makes more assumptions, and the model becomes unable to capture the important features of our dataset. A high bias model also cannot perform well on new data.
- ❖ Generally, a linear algorithm has a high bias, as it makes them learn fast.
- ❖ The simpler the algorithm, the higher the bias it has likely to be introduced. Whereas a nonlinear algorithm often has low bias.
- ❖ Some examples of machine learning algorithms with low bias are Decision Trees, k-Nearest Neighbours and Support Vector Machines.
- ❖ At the same time, an algorithm with high bias is Linear Regression, Linear Discriminant Analysis and Logistic Regression.

Ways to reduce High Bias:

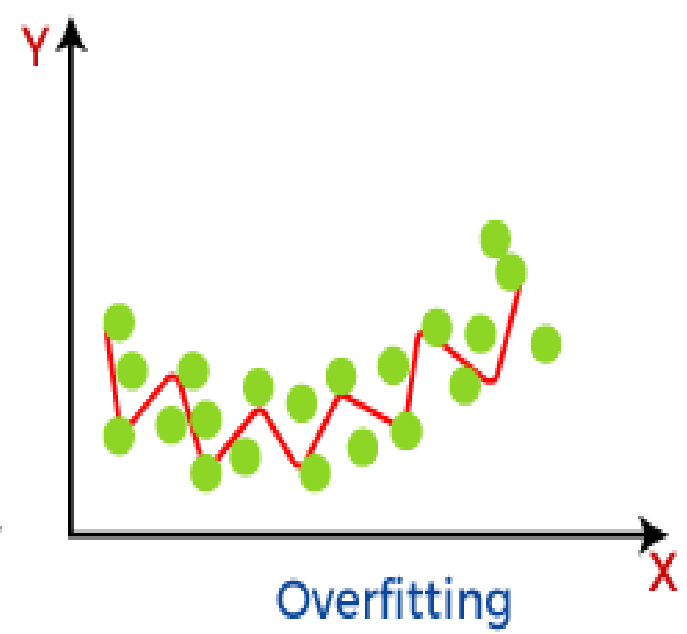
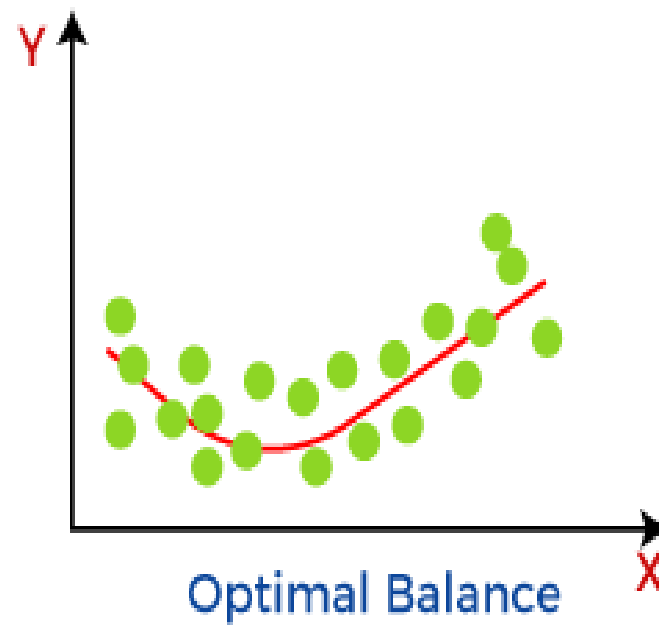
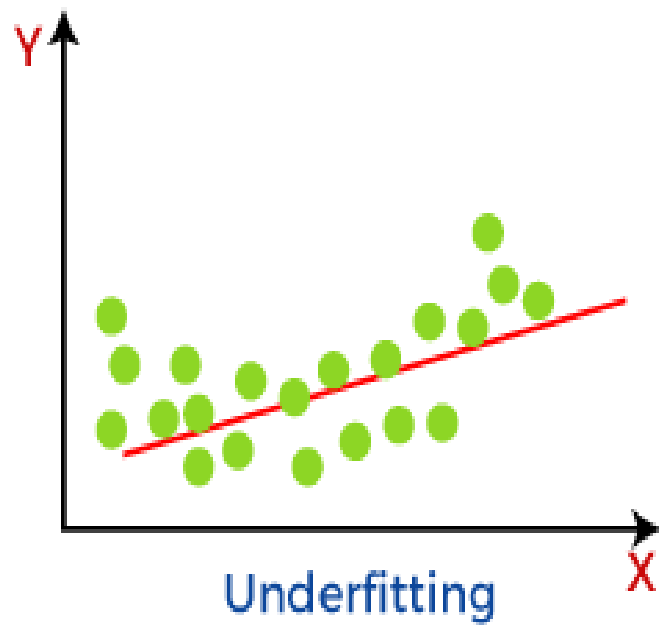
- ❖ High bias mainly occurs due to a much simple model. Below are some ways to reduce the high bias:
 - ❖ Identify potential sources of bias.
 - ❖ Identify accurate representative data
 - ❖ Increase the input features as the model is underfitted.
 - ❖ Decrease the regularization term.
 - ❖ Use more complex models, such as including some polynomial features.

Regularization is a technique used in machine learning and deep learning to prevent overfitting and improve the generalization performance of a model. It involves adding a penalty term to the loss function during training.

What is a Variance Error?

- ❖ The variance would specify the amount of variation in the prediction if the different training data was used.
- ❖ In simple words, variance tells that how much a random variable is different from its expected value.
- ❖ Ideally, a model should not vary too much from one training dataset to another, which means the algorithm should be good in understanding the hidden mapping between inputs and output variables.
- ❖ Variance errors are either of low variance or high variance.
- ❖ **How much the predicted target values from models trained using different samples vary from each other;** • **Noise: How much the sample target values differ from the true model.**

- ❖ **Low variance means there is a small variation in the prediction of the target function with changes in the training data set.**
- ❖ At the same time, High variance shows a large variation in the prediction of the target function with changes in the training dataset.
- ❖ A model that shows high variance learns a lot and perform well with the training dataset, and does not generalize well with the unseen dataset.
- ❖ As a result, such a model gives good results with the training dataset but shows high error rates on the test dataset.
- ❖ **Since, with high variance, the model learns too much from the dataset, it leads to overfitting of the model. A model with high variance has the below problems:**
 - ❖ **A high variance model leads to overfitting.**
 - ❖ **Increase model complexities.**
 - ❖ **Usually, nonlinear algorithms have a lot of flexibility to fit the model, have high variance.**



- ❖ Some examples of machine learning algorithms with low variance are, Linear Regression, Logistic Regression, and Linear discriminant analysis.
- ❖ At the same time, algorithms with high variance are decision tree, Support Vector Machine, and K-nearest neighbours.
 - ❖ Ways to Reduce High Variance:
 - ❖ Reduce the input features or number of parameters as a model is overfitted.
 - ❖ Do not use a much complex model.
 - ❖ Increase the training data.
 - ❖ Increase the Regularization term.

❖ **Different Combinations of Bias-Variance**

❖ **Low-Bias, Low-Variance:**

- ❖ The combination of low bias and low variance shows an ideal machine learning model. However, it is not possible practically.

❖ **Low-Bias, High-Variance:**

- ❖ With low bias and high variance, model predictions are inconsistent and accurate on average. This case occurs when the model learns with a large number of parameters and hence leads to an overfitting

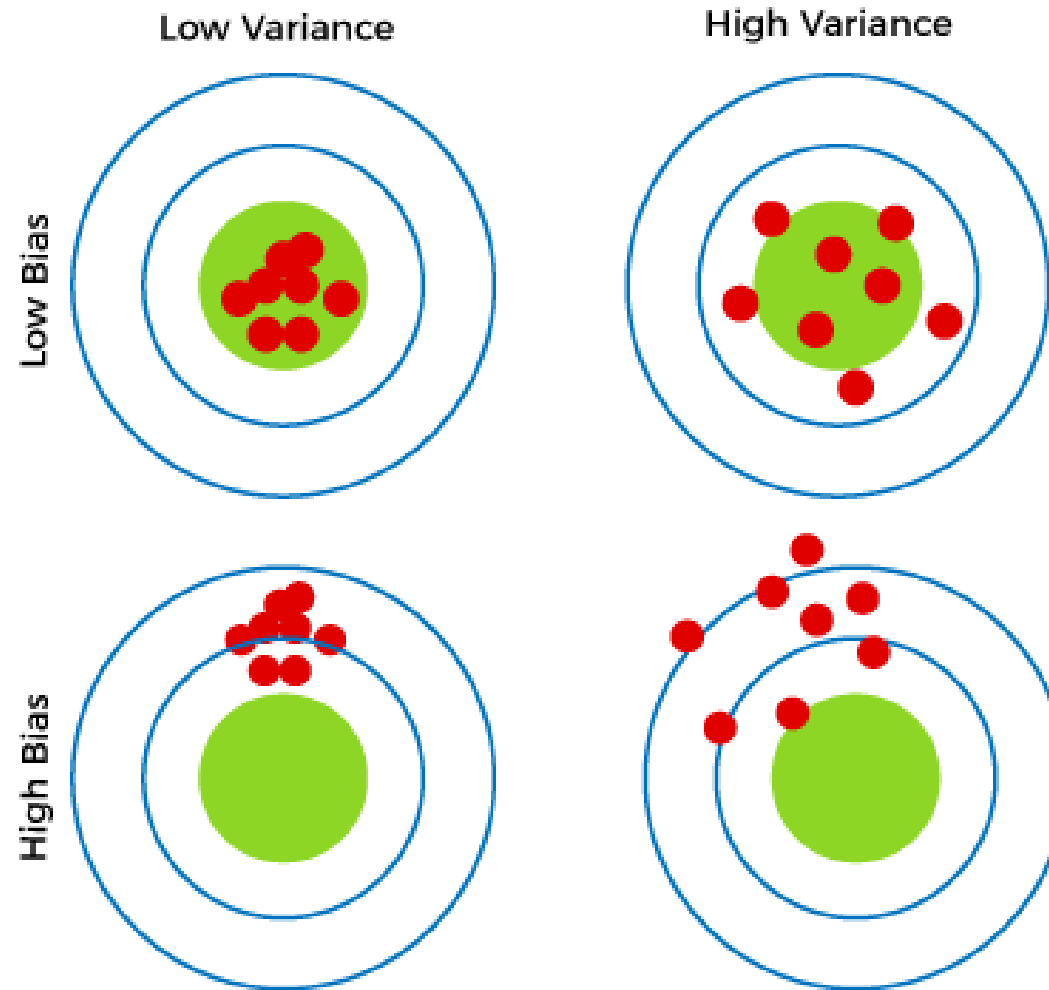
❖ **High-Bias, Low-Variance:**

- ❖ With High bias and low variance, predictions are consistent but inaccurate on average. This case occurs when a model does not learn well with the training dataset or uses few numbers of the parameter. It leads to underfitting problems in the model.

❖ **High-Bias, High-Variance:**

- ❖ With high bias and high variance, predictions are **inconsistent and also inaccurate** on average.

❖ There are four possible combinations of bias and variances, which are represented by the below diagram



❖ **Maximum Likelihood Estimation**

- ❖ The EM algorithm is considered a latent variable model to find the local maximum likelihood parameters of a statistical model, proposed by Arthur Dempster, Nan Laird, and Donald Rubin in 1977.
- ❖ **The EM (Expectation-Maximization) algorithm is one of the most commonly used terms in machine learning to obtain maximum likelihood estimates of variables that are sometimes observable and sometimes not observable .**
- ❖ However, it is also applicable to unobserved data or sometimes called latent (hidden).
- ❖ It has various real-world applications in statistics, including obtaining the mode of the **posterior marginal distribution** of parameters in machine learning and data mining applications.

Note:

Posterior: Conditional probability distribution representing what parameters are likely after observing the data object.

Likelihood: The probability of falling under a specific category or class.

- ❖ In most real-life time applications of machine learning, it is found that **several relevant learning features** are available, but very few of them are observable, and the rest are unobservable.
- ❖ If the variables are observable, then it can predict the value using instances.
- ❖ On the other hand, the variables which are latent (hidden) or directly not observable, for such variables Expectation-Maximization (EM) algorithm plays a vital role to predict the value with the condition that the general form of probability distribution governing those latent variables is known to us.

Application:

- ❖ A medical researcher is studying the relationship between a patient's age and whether they have a particular disease (yes/no). Using a dataset of patients, the researcher uses MLE to estimate the coefficients in a logistic regression model that predicts the probability of disease based on age.

❖ What is an EM algorithm?

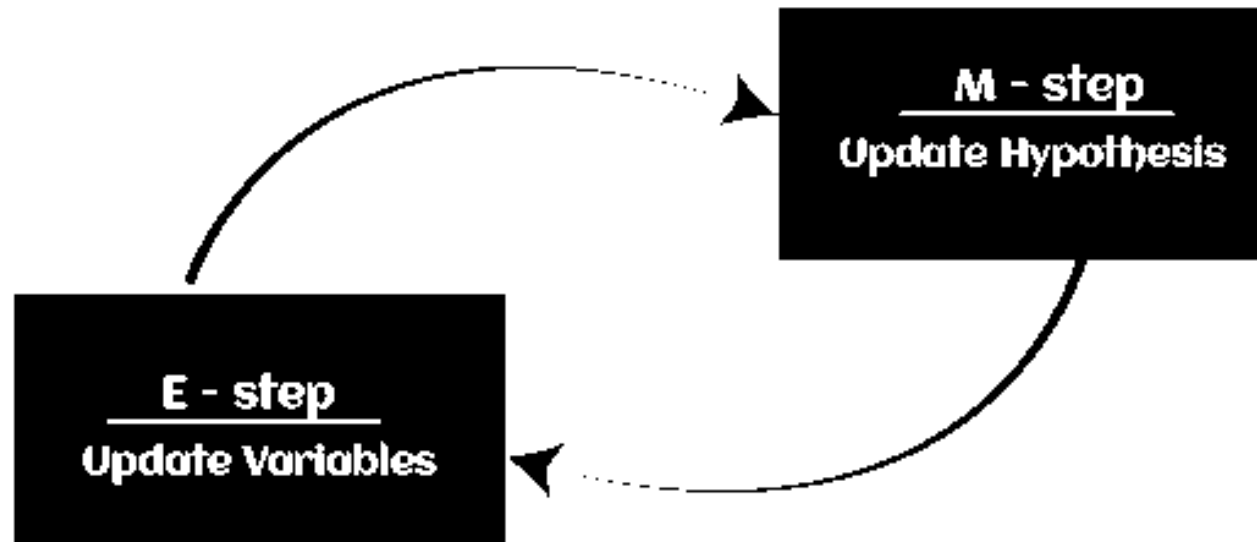
- ❖ The Expectation-Maximization (EM) algorithm is defined as the combination of various unsupervised machine learning algorithms, which is used to determine the **local maximum likelihood estimates** (MLE) or **maximum a posteriori estimates** (MAP) for unobservable variables in statistical models.
- ❖ Further, it is a technique to find maximum likelihood estimation when the latent variables are present. It is also referred to as the latent variable model.
- ❖ **A latent variable model consists of both observable and unobservable variables where observable can be predicted while unobserved are inferred from the observed variable. These unobservable variables are known as latent variables.**

Note:

- ❖ “A priori” and “a posteriori” refer primarily **to how, or on what basis, a proposition might be known.**
- ❖ **In general terms, a proposition is knowable a priori if it is knowable independently of experience, while a proposition knowable a posteriori is knowable on the basis of experience.**

❖ EM Algorithm

- ❖ The EM algorithm is the combination of various unsupervised ML algorithms, such as the **k-means clustering algorithm**. Being an iterative approach, it consists of two modes. In **the first mode**, we **estimate the missing or latent variables**. Hence it is referred to as the **Expectation/estimation step (E-step)**. Further, the **other mode** is used **to optimize the parameters of the models** so that it can explain the data more clearly. The second mode is known as the **maximization-step or M-step**.



- ❖ **Expectation step (E - step):** It involves the estimation (guess) of all missing values in the dataset so that after completing this step, there should not be any missing value.
- ❖ **Maximization step (M - step):** This step involves the use of estimated data in the E-step and updating the parameters.
- ❖ Repeat E-step and M-step until the convergence of the values occurs.
- ❖ **The primary goal of the EM algorithm is to use the available observed data of the dataset to estimate the missing data of the latent variables and then use that data to update the values of the parameters in the M-step.**

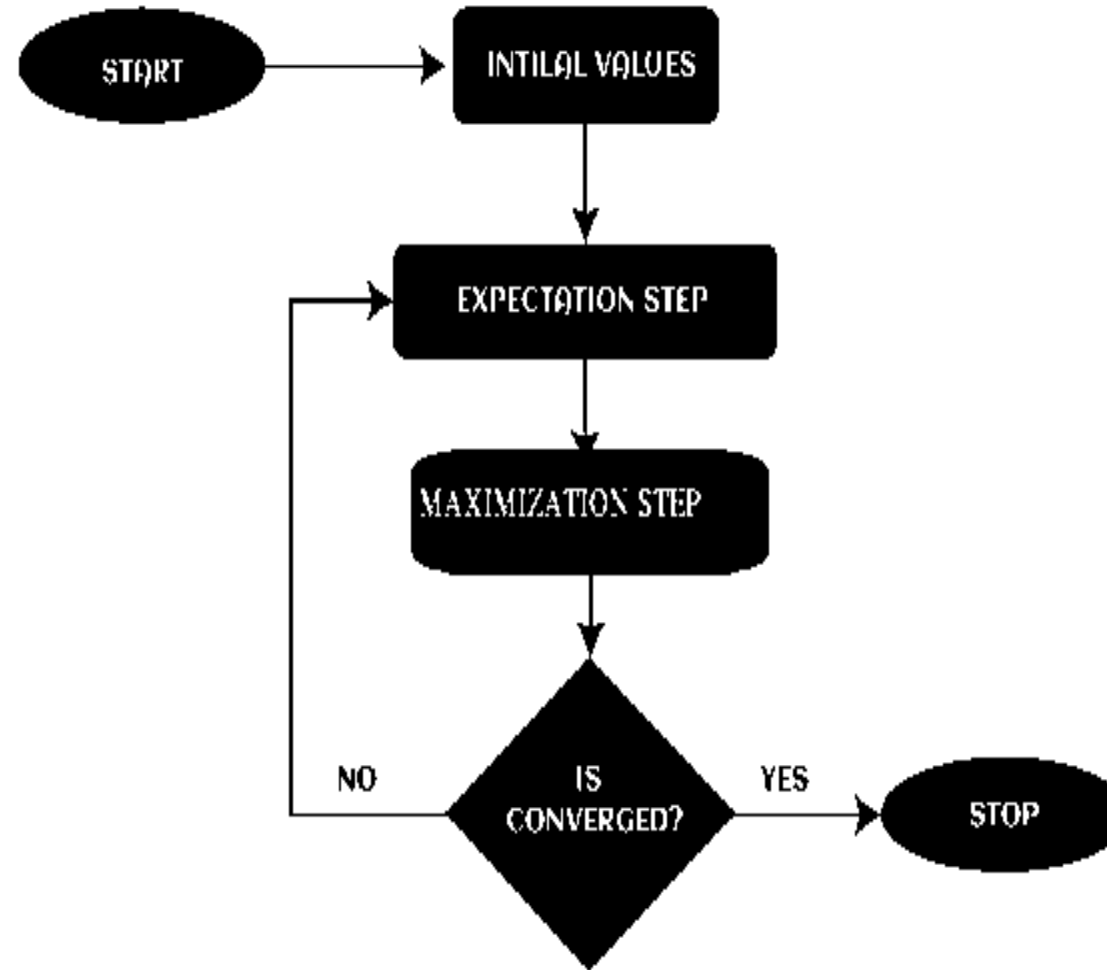
What is Convergence in the EM algorithm?

- ❖ Convergence is defined as the specific situation in probability based on intuition, e.g., if there are two random variables that have very less difference in their probability, then they are known as converged. In other words, whenever the values of given variables are matched with each other, it is called convergence.
- ❖ **Convergence in the Expectation-Maximization (EM) algorithm refers to the point at which the algorithm has reached a stable set of parameter estimates, meaning that further iterations will not significantly change these estimates.**

Steps in EM Algorithm

- ❖ The EM algorithm is completed mainly in 4 steps, which include Initialization Step, Expectation Step, Maximization Step, and Convergence Step. These steps are explained as follows:
- ❖ **Initialization Step :1st Step:** The very first step is to initialize the parameter values. Further, the system is provided with incomplete observed data with the assumption that data is obtained from a specific model.
- ❖ **Expectation Step 2nd Step:** This step is known as Expectation or E-Step, which is used to estimate or guess the values of the missing or incomplete data using the observed data. Further, E-step primarily updates the variables.
- ❖ **Maximization Step 3rd Step:** This step is known as Maximization or M-step, where we use complete data obtained from the 2nd step to update the parameter values. Further, M-step primarily updates the hypothesis.

❖ **Convergence Step 4th step:** The last step is to check if the values of latent variables are converging or not. If it gets "yes", then stop the process; else, repeat the process from step 2 until the convergence occurs.



❖ **Applications of EM algorithm**

- ❖ The primary aim of the EM algorithm is to estimate the missing data in the latent variables through observed data in datasets. The **EM algorithm or Latent Variable Model** has a broad range of real-life applications in machine learning. These are as follows:
- ❖ The EM algorithm is applicable in data clustering in machine learning.
- ❖ It is often used in computer vision and NLP (Natural language processing).
- ❖ It is used to estimate the value of the parameter in mixed models such as the Gaussian Mixture Model and quantitative genetics.
- ❖ It is also used in psychometrics for estimating item parameters and latent abilities of item response theory models.
- ❖ It is also applicable in the medical and healthcare industry, such as in image reconstruction and structural engineering.
- ❖ It is used to determine the Gaussian density of a function.

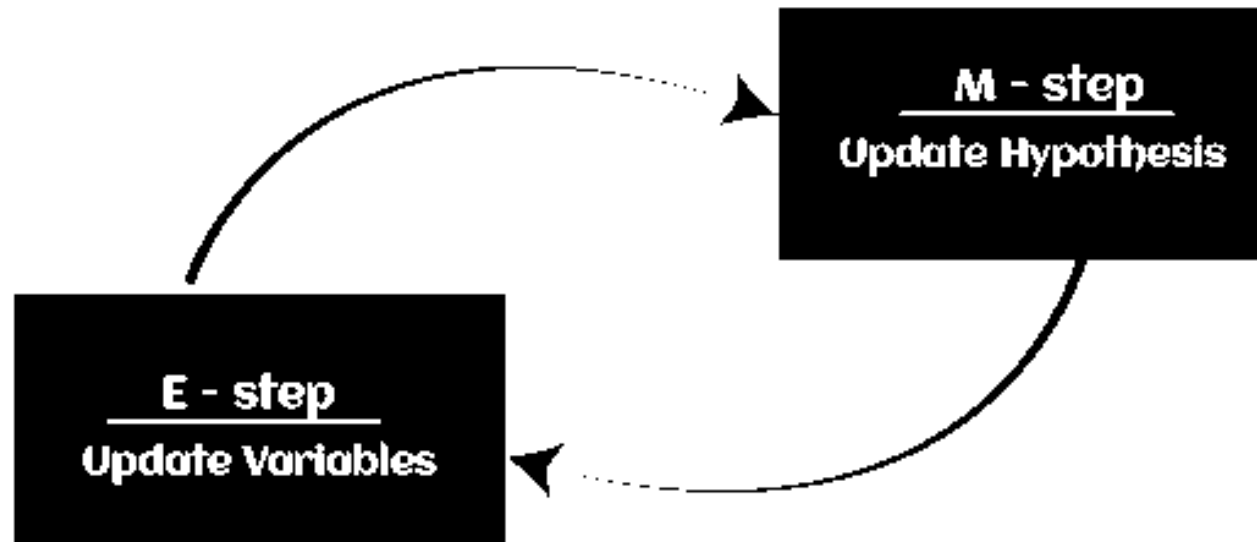
❑ Explore how MLE is used in classification tasks, such as object recognition or sentiment analysis, by deriving and implementing the MLE-based classifier

- ❖ Maximum Likelihood Estimation (MLE) is a statistical method used to find the parameters of a model that maximize the probability of observing the given data. In the context of classification tasks, MLE is used to determine the optimal model parameters that best fit the training data, allowing for accurate predictions on new, unseen data.
- ❖ **The MLE Process in Classification Model Selection:** Choose a suitable probability distribution for the data. This choice often depends on the nature of the classification problem (e.g., Bernoulli distribution for binary classification, multinomial distribution for multi-class classification).
- ❖ **Parameter Estimation:** Define the model parameters. These parameters represent the characteristics of the distribution that influence the classification decision.
- ❖ **Likelihood Function:** Construct the likelihood function, which expresses the probability of observing the given training data under the chosen model and parameters.

- ❖ **Maximization:** Find the values of the model parameters that maximize the likelihood function. This is typically done using optimization techniques like gradient ascent or Newton-Raphson.
- ❖ **Classification:** Once the optimal parameters are determined, the model can be used to classify new data points by calculating their likelihood under the learned model and assigning them to the class with the highest probability.
- ❖ **MLE in Object Recognition (Deep Learning)**
- ❖ In more complex tasks like object recognition, MLE is used in models like convolutional neural networks (CNNs). The process is conceptually similar:
- ❖ A CNN outputs a probability distribution over different classes (e.g., cat, dog, etc.) for a given input image.
- ❖ The probability distribution is based on the model's parameters (weights and biases).
- ❖ The likelihood is the probability that the CNN assigns to the correct label for each image.
- ❖ The log-likelihood is maximized during training using backpropagation and stochastic gradient descent to adjust the weights in the network.

❖ Implementing the MLE-based classifier.

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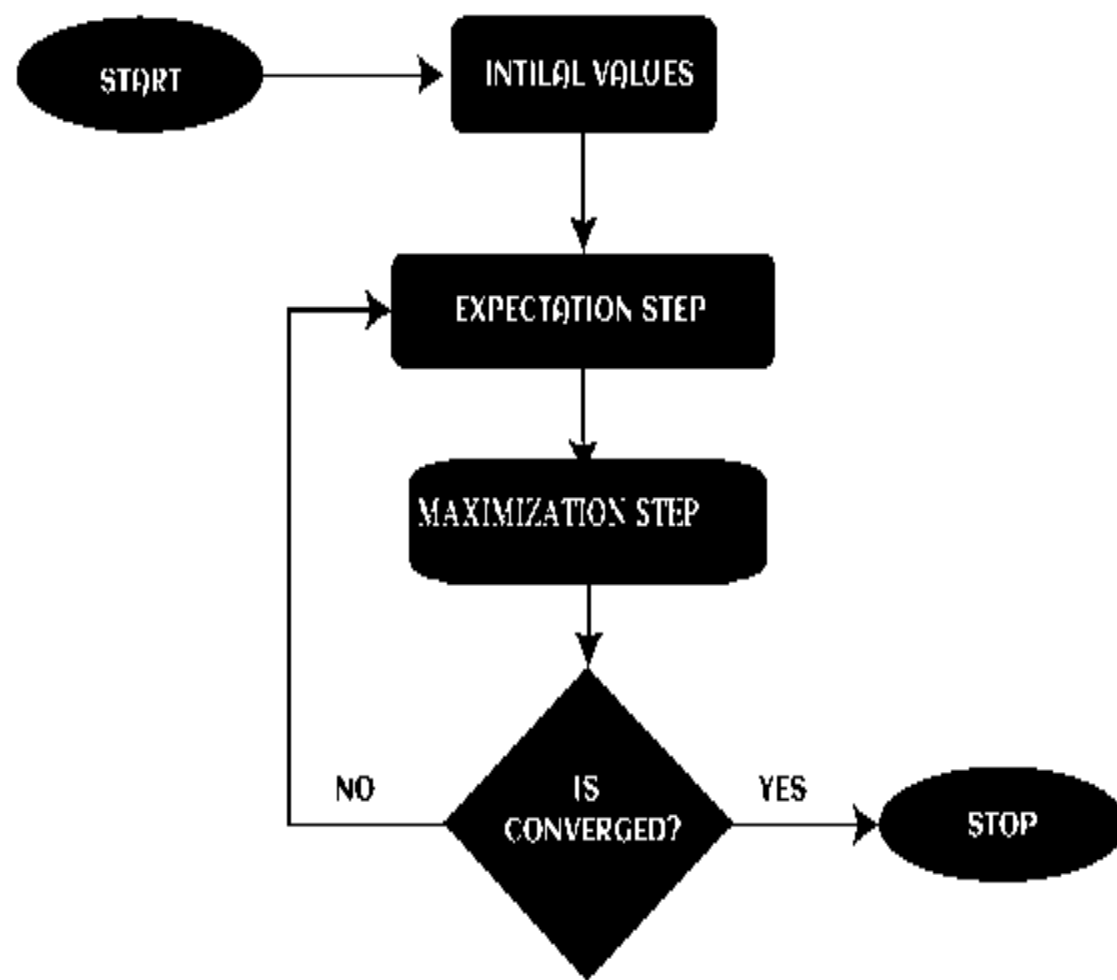


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Steps in EM Algorithm

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- ❖ **Convergence Step 4th step:** The last step is to check if the values of latent variables are converging or not. If it gets "yes", then stop the process; else, repeat the process from step 2 until the convergence occurs.



- ❖ In classification tasks, MLE is used to estimate model parameters that maximize the probability of correctly classifying the data.
- ❖ We construct a likelihood function based on the probability of the class labels given the data and model parameters.
- ❖ MLE maximizes the log-likelihood of observing the data by adjusting the parameters through optimization methods like gradient descent.

❖ The Basics of Bayesian Statistics

- ❖ Bayesian statistics is a branch of statistics that deals with uncertainty by incorporating prior knowledge or beliefs into the analysis of data.
- ❖ It is named after Thomas Bayes, an 18th-century **mathematician and theologian**. The fundamental concept in Bayesian statistics is the **Bayes' theorem**, which describes **how to update our beliefs in light of new evidence**. Here are the basics of Bayesian statistics:
- ❖ Bayesian statistics is a powerful approach for making statistical inferences, especially when dealing with small or complex data sets and when incorporating prior information is valuable or necessary.
- ❖ However, the choice of prior can be subjective and influence results, so it's essential to carefully consider and justify your prior assumptions in Bayesian analysis.

Bayesian statistics is a statistical theory and approach to data analysis that uses Bayes' theorem to describe the probability of an event based on previous knowledge and observed and unobserved parameters.

Bayes' Theorem

- ❖ **Bayes' Theorem**, named after 18th-century British mathematician Thomas Bayes, is a mathematical formula for determining conditional probability. **Conditional probability is the likelihood of an outcome occurring based on a previous outcome in similar circumstances.**
- ❖ The Bayes theorem is frequently referred to as the Bayes rule or Bayes Law. One of the most well-known theories in machine learning, the Bayes theorem helps determine the likelihood that one event will occur with unclear information while another has already happened.
- ❖ **Bayes' theorem describes the probability of occurrence of an event related to any condition. It is also considered for the case of conditional probability. Bayes theorem is also known as the formula for the probability of “causes”**

Conditional probability: Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Bayes' Theorem:

❖ At its core, Bayesian statistics is built upon Bayes' theorem, which relates conditional probabilities. The theorem can be expressed as follows:

$$P(A | B) = [P(B | A) * P(A)] / P(B)$$

- ❖ $P(A | B)$ represents the probability of event A occurring given that event B has occurred.
- ❖ $P(B | A)$ is the probability of event B occurring given that event A has occurred.
- ❖ $P(A)$ and $P(B)$ are the marginal probabilities of events A and B, respectively.
- ❖ In the context of Bayesian statistics:
 - ❖ $P(A | B)$ is the posterior probability, which represents our updated belief in A after observing B.
 - ❖ $P(B | A)$ is the likelihood, describing the probability of observing B if A is true.
 - ❖ $P(A)$ is the prior probability, representing our initial belief in A before considering B.
 - ❖ $P(B)$ is the marginal likelihood or evidence, the probability of observing B without considering A.

❖ **Prior Probability:**

- ❖ In Bayesian statistics, prior knowledge or beliefs about the probability of an event play a crucial role. This is known as the prior probability ($P(A)$), which represents your initial beliefs about the likelihood of an event occurring before observing any data. The choice of the prior can significantly influence the final results.

❖ **Likelihood:**

- ❖ The likelihood ($P(B \mid A)$) represents the probability of observing the data (B) given a specific hypothesis or model (A). It quantifies how well the hypothesis or model explains the observed data. It's essential to choose an appropriate likelihood function that accurately reflects the relationship between your data and the parameters you want to estimate.

❖ **Posterior Probability:**

- ❖ The posterior probability ($P(A | B)$) is the updated belief in the hypothesis or model A after observing the data B. It is calculated using Bayes' theorem and combines the prior knowledge and the likelihood of the data. The posterior probability provides a more informed and updated estimate of the event's probability based on the observed data.

❖ **Bayesian Inference:**

- ❖ Bayesian inference involves using Bayes' theorem to estimate or update the parameters of a statistical model based on observed data. This process typically includes:
 - ❖ Choosing a prior distribution that reflects your initial beliefs.
 - ❖ Defining a likelihood function that describes the data-generation process.
 - ❖ Applying Bayes' theorem to calculate the posterior distribution, which represents updated parameter estimates based on the data.

❖ **Supervised Learning and Unsupervised Learning algorithms:**

Aspect	Supervised Learning	Unsupervised Learning
Objective	Predict a target variable or label based on input data.	Discover patterns, structures, or relationships in the data without a specific target variable.
Training Data	Labeled data with input-output pairs.	Unlabeled data with only input features.
Learning Process	The algorithm learns by comparing predictions to actual labels and adjusting model parameters to minimize errors.	The algorithm explores the data's inherent structure, often without predefined categories or labels.
Output	Provides predictions or classifications for new, unseen data.	Typically produces clusters or representations of data points that share similarities or patterns.
Examples	Classification, Regression.	Clustering, Dimensionality Reduction, Association Rule Mining.

Aspect	Supervised Learning	Unsupervised Learning
Evaluation Metrics	Commonly uses metrics like accuracy, mean squared error (MSE), precision, recall, F1-score, etc.	Evaluation metrics depend on the specific task, such as silhouette score for clustering, explained variance for dimensionality reduction, etc.
Use Cases	Used for tasks where there is a clear target variable to predict, e.g., image classification, spam detection, stock price prediction.	Applied in situations where you want to explore and uncover hidden patterns or group similar data points, e.g., customer segmentation, anomaly detection, topic modeling.
Example Algorithms	Linear Regression, Support Vector Machines (SVM), Decision Trees, Neural Networks.	K-Means Clustering, Principal Component Analysis (PCA), Association Rules, Hierarchical Clustering.
Supervision Level	High supervision, as the algorithm relies on labeled data for training.	Low or no supervision, as it works with unlabeled data or minimal guidance.

Aspect	Supervised Learning	Unsupervised Learning
Data Requirement	Requires a significant amount of labeled data for training.	Can work with both labeled and unlabeled data, making it more versatile in some scenarios.
Common Challenges	Overfitting, data imbalance, labeling bias.	Determining the appropriate number of clusters, noise handling, and interpretation of discovered patterns.

❖ Building a Machine Learning Algorithm

- ❖ Nearly all deep learning algorithms can be described as particular instances of a fairly simple recipe: **combine a specification of a dataset, a cost function, an optimization procedure** and a model.
- ❖ For example, the linear regression algorithm combines a dataset consisting of X and y , the cost function.

$$J(w, b) = -\mathbb{E}_{x, y \sim \hat{p}_{\text{data}}} \log p_{\text{model}}(y \mid x),$$

- ❖ the model specification $p_{\text{model}}(y \mid x) = \mathcal{N}(y; x^\top w + b, 1)$, and, in most cases, the optimization algorithm defined by solving for where the **gradient of the cost (slope)** is zero using the normal equations.
- ❖ By realizing that we can replace any of these components mostly independently from the others, we can obtain a very wide variety of algorithms.
- ❖ The **cost function** typically includes at least one term that causes the learning process to perform statistical estimation. **The most common cost function is the negative log-likelihood, so that minimizing the cost function causes maximum likelihood estimation.**

- ❖ The cost function may also include additional terms, such as **regularization terms**. For example, we can **add weight** decay to the linear regression cost function to obtain.

$$J(w, b) = \lambda \|w\|_2^2 - \mathbb{E}_{x, y \sim p_{\text{data}}} \log p_{\text{model}}(y | x).$$

- ❖ This still allows closed-form optimization.
- ❖ If we change the model to be nonlinear, then most cost functions can no longer be optimized in closed form. This requires us to choose an iterative numerical optimization procedure, such as gradient descent.
- ❖ The recipe for **constructing a learning algorithm by combining models, costs, and optimization algorithms supports** both supervised and unsupervised learning.
- ❖ The **linear regression** example shows how to support supervised learning.
- ❖ Unsupervised learning can be supported by defining a dataset that contains only X and providing an appropriate unsupervised cost and model.

❖ For example, we can obtain the first PCA vector by specifying that our loss function is

$$J(\mathbf{w}) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \|\mathbf{x} - r(\mathbf{x}; \mathbf{w})\|_2^2$$

❖ while our model is defined to have $\mathbf{w}$ with norm one and reconstruction function

$$r(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} \mathbf{w}.$$

- ❖ In some cases, the cost function may be a function that we cannot actually evaluate, for computational reasons.
- ❖ In these cases, we can still approximately minimize it using iterative numerical optimization so long as we have some way of approximating its gradients.
- ❖ Most machine learning algorithms make use of this recipe, though it may not immediately be obvious.

- ❖ If a machine learning algorithm seems especially unique or hand-designed, it can usually be understood as using a special-case optimizer.
- ❖ Some models such as **decision trees or k-means** require special-case optimizers because their cost functions have flat regions that make them inappropriate for minimization by gradient-based optimizers.
- ❖ Recognizing that most machine learning algorithms can be described using this recipe helps to see the different algorithms as part of a taxonomy of methods for doing related tasks that work for similar reasons, rather than as a long list of algorithms that each have separate justifications

❖ **Challenges and Motivation to Deep learning.**

- ❖ **During the development phase our focus is to select a learning algorithm and train it on some data, the two things that might be a problem are a bad algorithm or bad data, or perhaps both of them.**
- ❖ Many machine learning problems become exceedingly difficult when the number of dimensions in the data is high. This phenomenon is known as the curse of dimensionality. Of particular concern is that the number of possible distinct configurations of a set of variables increases exponentially as the number of variables increases.

1. Not enough training data :

- ❖ Let's say for a child, to make him learn what an apple is, all it takes for you to point to an apple and say apple repeatedly.
- ❖ Now the child can recognize all sorts of apples.
- ❖ Well, machine learning is still not up to that level yet; it takes a lot of data for most of the algorithms to function properly.
- ❖ For a simple task, it needs thousands of examples to make something out of it, and for advanced tasks like image or speech recognition, it may need lakhs(millions) of examples

2. Poor Quality of data:

- ❖ Obviously, if your training data has lots of errors, outliers, and noise, it will make it impossible for your machine learning model to detect a proper underlying pattern. Hence, it will not perform well.
- ❖ So put in every ounce of effort in cleaning up your training data. No matter how good you are in selecting and hyper tuning the model, this part plays a major role in helping us make an accurate machine learning model.
- ❖ **“Most Data Scientists spend a significant part of their time in cleaning data”.**
- ❖ There are a couple of examples when you'd want to clean up the data :
- ❖ If you see some of the instances are clear outliers just discard them or fix them manually.
- ❖ If some of the instances are missing a feature like (E.g., 2% of user did not specify their age), you can either ignore these instances, or fill the missing values by median age, or train one model with the feature and train one without it to come up with a conclusion

3. Irrelevant Features:

- ❖ “Garbage in, garbage out (GIGO).”
- ❖ Features are relevant if they are either strongly or weakly relevant, and are irrelevant otherwise.
- ❖ Irrelevant features can never contribute to prediction accuracy, by definition.
- ❖ The credit for a successful machine learning project goes to coming up with a good set of features on which it has been trained (often referred to as feature engineering), which includes feature selection, extraction, and creating new features which are other interesting topics to be covered in upcoming blogs.

4. Non-representative training data:

- ❖ To make sure that our model generalizes well, we have to make sure that our training data should be representative of the new cases that we want to generalize to.
- ❖ If train our model by using a non-representative training set, it won't be accurate in predictions it will be biased against one class or a group.
- ❖ For E.G., Let us say you are trying to build a model that recognizes the genre of music. One way to build your training set is to search it on YouTube and use the resulting data.
- ❖ Here we assume that YouTube's search engine is providing representative data but in reality, the search will be biased towards popular artists and maybe even the artists that are popular in your location(if you live in India you will be getting the music of SBP, etc).
- ❖ So use representative data during training, so your model won't be biased among one or two classes when it works on testing data.

5. Overfitting and Underfitting :

- ❖ **Overfitting** is an undesirable machine learning behavior that occurs when the machine learning model gives accurate predictions for training data but not for new data. When data scientists use machine learning models for making predictions, they first train the model on a known data set.
- ❖ Overfitting happens when our model is too complex.
- ❖ Things which we can do to overcome this problem:
 - ❖ Simplify the model by selecting one with fewer parameters.
 - ❖ By reducing the number of attributes in training data.
 - ❖ Constraining the model.
 - ❖ Gather more training data.
 - ❖ Reduce the noise.

❖ **Underfitting**

- ❖ When a model has not learned the patterns in the training data well and is unable to generalize well on the new data, it is known as underfitting.
- ❖ An underfit model has poor performance on the training data and will result in unreliable predictions.
- ❖ Things which we can do to overcome this problem:
 - ❖ Select a more advanced model, one with more parameters.
 - ❖ Train on better and relevant features.
 - ❖ Reduce the constraints.

All of us do not have
equal talent.

But , all of us have an
equal opportunity to
develop our talents.

A. P. J. Abdul Kalam



A red pushpin is pinned to the top center of a yellow sticky note.

Thank
You!